Aspirin: Mechanism of Action

The pharmacological activity of aspirin derives principally from its irreversible inhibition of the cyclooxygenase enzymes, designated COX-1 and COX-2, which catalyse the conversion of arachidonic acid into prostaglandin H2. This intermediate serves as the common precursor for the prostaglandins, prostacyclin, and the thromboxanes. Aspirin is distinguished from the majority of nonsteroidal anti-inflammatory drugs by the covalent nature of this inhibition: an acetyl group is transferred from the drug to a serine residue within the enzyme's active channel, sterically obstructing the entry of substrate. Because the modification is permanent, restoration of enzymatic function requires synthesis of new enzyme.

This mechanism accounts for the unusual pharmacodynamic profile of aspirin in platelets. Mature platelets are anucleate and therefore incapable of replacing acetylated COX-1, with the consequence that a single dose suppresses thromboxane A2 production for the remaining lifespan of the affected cell, ordinarily between seven and ten days. Recovery of platelet function at the population level proceeds only as the bone marrow releases new platelets into circulation. The clinical implication is that antiplatelet efficacy is determined less by plasma concentration than by cumulative exposure, which explains why doses substantially below those required for analgesia remain fully effective for cardiovascular prophylaxis.

The differential sensitivity of COX-1 and COX-2 to aspirin underlies the concept of a therapeutic window for antiplatelet dosing. Endothelial cells retain the capacity for protein synthesis and can therefore replenish COX-2 and continue producing prostacyclin, a vasodilator and inhibitor of platelet aggregation. At low doses, presystemic acetylation of platelet COX-1 in the portal circulation predominates while endothelial prostacyclin output is comparatively preserved, yielding a favourable balance between antithrombotic and prothrombotic mediators. Higher doses erode this selectivity.

Aspirin does not merely silence COX-2 but redirects its catalytic behaviour. The acetylated enzyme retains oxygenase activity while losing its cyclooxygenase function, generating 15-R-hydroxyeicosatetraenoic acid, which is subsequently converted by leukocyte 5-lipoxygenase into a class of compounds termed aspirin-triggered lipoxins. These mediators actively promote the resolution of inflammation rather than simply suppressing its initiation, a distinction that has attracted considerable interest in the study of specialised pro-resolving mediators.

Additional mechanisms have been proposed but remain less firmly established. Salicylate, the principal metabolite, has been reported to activate AMP-activated protein kinase and to modulate signalling through the transcription factor NF-kappa-B, effects that occur at concentrations achieved with anti-inflammatory rather than antiplatelet dosing. Aspirin also uncouples oxidative phosphorylation at high concentrations, which may explain the paradoxical hyperthermia observed in salicylate overdose.